



Neuromarkers in addiction: definitions, development strategies, and recent advances

Nicholas R. Harp¹ · Tor D. Wager² · Hedy Kober¹

Received: 21 November 2023 / Accepted: 12 March 2024

© The Author(s), under exclusive licence to Springer-Verlag GmbH Austria, part of Springer Nature 2024

Abstract

Substance use disorders (SUDs) are the most costly and prevalent psychiatric conditions. Recent calls emphasize a need for biomarkers—measurable, stable indicators of normal and abnormal processes and response to treatment or environmental agents—and, in particular, brain-based neuromarkers that will advance understanding of the neurobiological basis of SUDs and clinical practice. To develop neuromarkers, researchers must be grounded in evidence that a putative marker (i) is sensitive and specific to the psychological phenomenon of interest, (ii) constitutes a predictive model, and (iii) generalizes to novel observations (e.g., through internal cross-validation *and* external application to novel data). These neuromarkers may be used to index risk of developing SUDs (susceptibility), classify individuals with SUDs (diagnostic), assess risk for progression to more severe pathology (prognostic) or index current severity of pathology (monitoring), detect response to treatment (response), and predict individualized treatment outcomes (predictive). Here, we outline guidelines for developing and assessing neuromarkers, we then review recent advances toward neuromarkers in addiction neuroscience centering our discussion around neuromarkers of craving—a core feature of SUDs. In doing so, we specifically focus on the Neurobiological Craving Signature (NCS), which show great promise for meeting the demand of neuromarkers.

Keywords Biomarker · Neuromarker · Craving · Substance use disorder · Machine learning

Introduction

Substance use disorders (SUDs) are the most deadly and costly psychiatric disorders. Indeed, healthcare and lost productivity costs from cigarette smoking alone exceed \$600 billion annually in the U.S. (CDC 2022), and costs across misused substances more broadly exceed \$2 trillion (CDC 2019, 2022; Luo et al. 2021; Nicosia et al. 2009). Across drug types, SUDs manifest as problematic patterns of substance use, driven in large part by intense urges to consume the drug (i.e., cravings; Vafaie and Kober 2022). Such use culminates in consumption of larger amounts or for longer periods than intended (i.e., risky use, impaired control) that may lead to pharmacological tolerance, and ultimately

adversely impacts non-drug goals (e.g., workplace, social, health; American Psychiatric Association 2013).

Substances abused by humans have diverse primary mechanisms of action (for a review, see Volkow et al. 2019). Indeed, numerous neurotransmitter receptors known today bear their namesake from the exogenous substances that bind to and stimulate them. For instance, the ionotropic nicotinic acetylcholine receptor is distinguished from the metabotropic muscarinic acetylcholine receptor (to which muscarine, but not nicotine, binds; Schleifer and Eldefrawi 1974). Similarly, researchers established that cannabis and opiate drugs bind to cannabinoid (Devane et al. 1988) and opioid receptors (Simon et al. 1973), respectively, prior to the discovery of the endogenous cannabinoids (e.g., anandamide; Devane et al. 1992) and opioid peptides (e.g., enkephalins; Hughes et al. 1975). Other drugs, like cocaine, block the reuptake of dopamine into cell bodies (Ritz et al. 1987), and methamphetamine stimulates the release of monoamines (e.g., dopamine, norepinephrine) and reverses the monoamine transporters to flood the synapse with these neurotransmitters (Sulzer et al. 2005). Still, alcohol elicits a relatively complex series of effects, including allosteric

✉ Hedy Kober
hedy.kober@yale.edu

¹ Department of Psychiatry, Yale University, New Haven, CT, USA

² Department of Psychological & Brain Sciences, Dartmouth College, Hanover, NH, USA

enhancement of GABA_A receptor activity (Mihic et al. 1997) and inhibition of glutamatergic N-methyl D-aspartate receptors (Lovinger et al. 1989), among others (Volkow et al. 2019).

That said, these same substances share similarities in their secondary effects; that is, substances abused by humans typically engage the mesolimbic dopamine pathway (Di Chiara and Imperato 1988; Nestler 2005), along with other neurochemical systems (e.g., opioid peptide, endocannabinoid signaling; Herz 1997; Koob and Volkow 2016; Le Merrer et al. 2009; Massaly et al. 2016; Peters et al. 2021). Substances of abuse also induce long-term neuroplastic changes in these systems (Berridge and Robinson 2016; Lüscher and Malenka 2011; Mameli and Lüscher 2011), studied primarily in animal models of drug exposure and use (Ahmed 2010). In humans, neuroimaging data collected via functional magnetic resonance imaging (fMRI) suggests differences in the brain response of individuals with SUDs, relative to controls, during the presentation of various stimuli, including drug-relevant cues (e.g., Garavan et al. 2000; Jasinska et al. 2014; Schacht et al. 2013). Thus, data converging from both animal and human studies over the last few decades demonstrate a clear neurobiological basis for SUDs (Heilig et al. 2021; Koob and Volkow 2016; Leshner 1997) with some drug-specific differences (e.g., primary mechanisms of action), and also many commonalities (e.g., action on dopamine, craving, other SUD symptoms). Nonetheless, there remain challenges to identifying neurobiological signatures—e.g., patterns of fMRI activity across brain regions—of addiction in humans and of its constituent processes (e.g., craving). In this context, the application of computational methods to neuroimaging data is showing increasing promise.

Computational psychiatry is a subfield that aims to leverage both data- and theory-driven approaches to understand the neurobiological basis of psychiatric disorders and to advance clinical practice (Huys et al. 2016; Wang and Krystal 2014). Computational psychiatry applies formal mathematical models to understand adaptive and maladaptive processes (e.g., learning and decision-making; Collins and Shenhav 2022), in a manner that informs the underlying neural computations. Broadly, computational approaches either characterize dynamic changes over time (e.g., reinforcement learning models) or recognize and extract patterns from high dimensional, multivariate data. The two approaches are complementary. For example, reinforcement learning models provide much evidence that decision-making processes are altered in addiction, likely via compromised computations in the mesocorticolimbic dopamine pathway (Mollick and Kober 2020, for review). Indeed, the ventromedial prefrontal cortex (vmPFC), orbitofrontal cortex, and anterior cingulate cortex—downstream targets of the corticostriatal pathway—display distinct multivariate patterns of trial-by-trial activity

during reward prediction errors, and these dynamics are statistically associated with subsequent changes in behavioral responses (Möhring and Gläscher 2023). Still, such approaches remain insufficient for a major challenge facing computational psychiatry: translation from computation to the clinic (Yip et al. 2022). Certainly, scientific advances from computational psychiatry, including the application of computational methods to neuroimaging, are not routinely used in clinical practice—at least in part, owing to a focus on narrow processes (e.g., reinforcement learning parameters) rather than psychological states that represent psychiatric symptoms (e.g., craving in SUDs).

To address this gap, we believe that researchers can leverage computational methods, including machine learning and artificial intelligence, and combine them with high dimensional neuroimaging data to characterize the brain basis of component processes or symptoms of psychiatric disorders in pursuit of biomarkers. Biomarkers are *measurable, stable biological indicators* of either normal or abnormal processes and responses to intervention/treatment or environmental agents (FDA-NIH Biomarker Working Group 2016). Importantly, biomarkers are distinct from other biological patterns associated with disease in that they are optimized to make accurate predictions about new data. Calls for the development of biomarkers for psychiatric disorders, and SUDs specifically, are increasingly common (Davis et al. 2020; Volkow and Baler 2013; Volkow and Boyle 2018; Volkow et al. 2015; Woo et al. 2017). Nonetheless, despite a clear trend toward increasing reports of neuromarkers derived from human neuroimaging, SUDs and addiction have been largely overlooked (see Fig. 1).

We refer to brain-based biomarkers as *neuromarkers*, and we ground our discussion in such neuromarkers (note that the terms are largely interchangeable, with neuromarkers representing a specific, brain-based instance of the broader term biomarkers). Objective neuromarkers of disease state and response to treatment promise to overcome numerous limitations in psychiatry, including the lack of neurobiological indicators, or a brain basis, for psychiatric nosology (American Psychiatric Association 2013), impaired patient insight into disease state and treatment response of psychiatric disorders (Goldstein et al. 2009), and challenges in identifying personalized, optimal treatment strategies (Volkow and Baler 2013). Pending adequate validation (described in detail below), neuromarkers could eventually become valuable clinical and research tools. For instance, a well-validated diagnostic neuromarker that distinguishes individuals with versus without SUDs could assist researchers in selecting study participants, and could assist clinicians in identifying individuals in need of initiating treatment, and even provide a basis for insurance reimbursement (treatment cost is a notable barrier for individuals living with

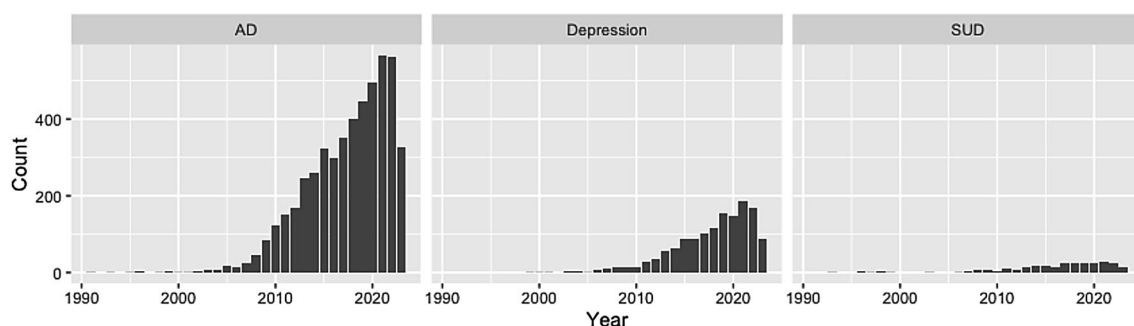


Fig. 1 PubMed search results for “biomarker” AND “neuroimaging” AND [disease keyword] as of June 30, 2023. Year is shown on the x-axis and publication count on the y-axis. *AD* Alzheimer’s Disease, *SUD* substance use disorder

SUDs; Dickson-Gomez et al. 2022). Similarly, a well-validated predictive neuromarker could match individuals to treatment given their neurobiology in both research and clinical settings. Finally, a well-validated prognostic neuromarker, predictive of post-treatment use severity or relapse likelihood, could inform prognosis following treatment (Fig. 2). Still, the specific role of any biomarker will depend upon the evidence base for its validation in well-defined use cases. Here, we review recent advances in the field of translational human neuroimaging and addiction neuroscience, by (1) defining neuromarkers and identifying prerequisites for their development and assessment, (2) integrating a common understanding of neuromarkers and their purpose(s) with recent efforts and examples, and (3) looking toward the future of neuromarker development in addiction neuroscience.

Defining, developing, and assessing neuromarkers

As the field seeks to identify, validate, and ultimately translate biomarkers into the clinic, it will be crucial to precisely define biomarkers and their potential uses (FDA-NIH Biomarker Working Group 2016), both for the scientists who develop biomarkers and the regulatory agencies that monitor the safety of medical products (Califf 2018). To promote “effective, unambiguous communication,” the Food and Drug Administration (FDA) and National Institutes of Health (NIH) released the Biomarkers EndpointS and other Tools (BEST) resource to define biomarkers and their purpose (FDA-NIH Biomarker Working Group 2016). Following these BEST guidelines, biomarkers can range from specific analytes (e.g., drug metabolites) and anatomical features to systems-level physiological characteristics (e.g.,

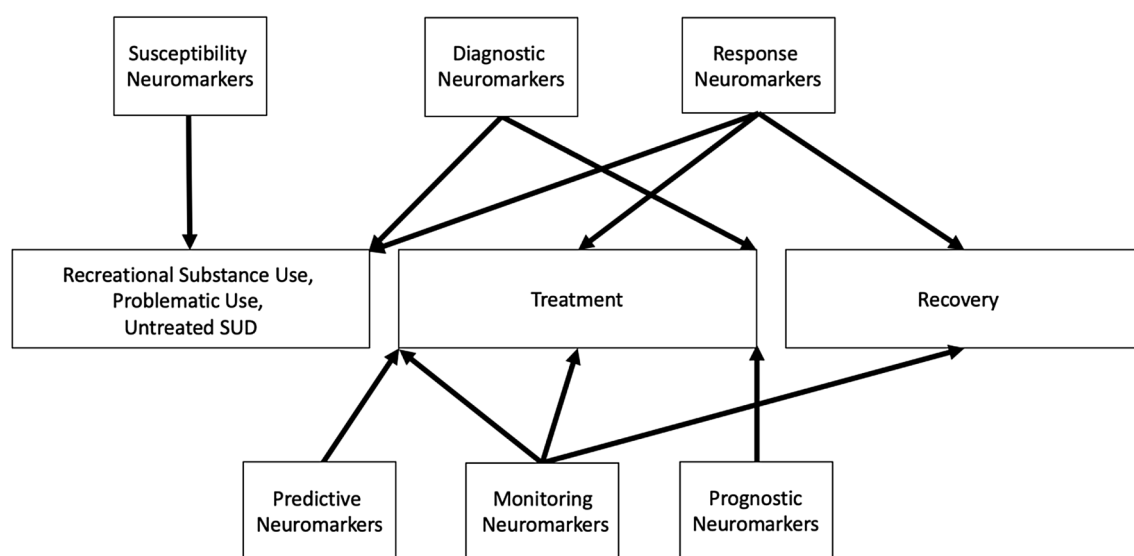


Fig. 2 A non-prescriptive model of possible neuromarker use cases in clinical practice

brain activity/connectivity). These markers can index normal or abnormal processes, as well as responses to interventions or environmental agents (FDA-NIH Biomarker Working Group 2016).

Ultimately, a neuromarker links brain-based measures to disease outcomes through a biologically plausible pathway. For instance, decades of human neuroimaging research, as well as more invasive experimental neurobiology and non-human animal research (Berridge and Robinson 2016; Contreras et al. 2007), suggest a central role for regions including the amygdala, insula, and striatum in SUDs and addiction (Di Chiara and Imperato 1988; Koob and Volkow 2016). Likewise, neuroimaging meta-analyses, which interrogate activation across diverse methodological choices (e.g., stimuli, task; Ekhtiari et al. 2022), can also provide insight into the plausibility of potential neuromarkers by uncovering core circuits consistently activated in response to specific events (e.g., drug cue presentation; Chase et al. 2011; Engelmann et al. 2012; Hill-Bowen et al. 2021; Noori et al. 2016; Schacht et al. 2013). As such, neuromarkers for SUDs and other addictive disorders will likely capitalize on activity patterns in many of these previously identified regions. Upon identification of a plausible biological pathway, researchers can develop neuromarkers that provide insight into disease/disorder susceptibility, progression, and response to treatment or intervention (ACRI Network et al., 2024). Appropriate clinical use will necessarily vary according to the purpose of the neuromarker (more in the following section). Regardless of use case, a shared, common understanding of neuromarkers' definition and purpose—as well as how to appropriately develop and assess neuromarkers—are crucial for making progress toward implementation in clinical practice. Below, we review some prerequisites for neuromarker development and strategies for assessing neuromarker performance.

Sensitivity and specificity

In our view, a neuromarker must be grounded in evidence for both sensitivity and specificity of the phenomenon of interest (e.g., psychological states, case vs. control), and also show predictive utility and out-of-sample generalization (i.e., new individuals, different tasks/paradigms and stimulus modalities, etc.). We also expect neuromarkers to achieve “benchmark” rates of sensitivity and specificity, which refer to accurate classification of the phenomenon of interest—whether that be psychological states (e.g., accurately classifying moments an individual experiences as craving as “craving” and other moments—like moments of pain or anger—as “not craving”), or persons (i.e., accurately classifying individuals with the disease/pathology from individuals without the disease/pathology). Notably, positive predictive values (i.e., the proportion of positive

results that are truly positive; PPV) are influenced by the base rate of the phenomenon (Poldrack et al. 2020). Thus, assessing sensitivity and specificity (and determining appropriate “benchmarks”) requires consideration of the costs of both incorrect positive and negative predictive values (e.g., clinical or societal costs of incorrect classification; Woo et al. 2017). In turn, this means that acceptable “benchmark” levels of sensitivity and specificity will be field- and case-dependent. The sensitivity and specificity of diagnostic and other classification neuromarkers (discussed below) should be assessed using receiver operating characteristic curves, which characterize performance across a range of criterion thresholds. It is important to note that all markers have and will fall short of perfect performance (FDA-NIH Biomarker Working Group 2016). Still, upon obtaining sufficient sensitivity and specificity, researchers can then proceed to establish the predictive utility of the neuromarker in novel data and assess its generalizability to provide greater confidence in its viability as a neuromarker.

Unfortunately, confusion about the definition and requirements for establishing neuro- and biomarkers pervades much of the human neuroimaging literature. Namely, previous reports of “potential biomarkers” have often reflected statistical associations within a single sample and were not evaluated to the rigorous degree required for establishment as a biomarker. For instance, consider the recent finding that the degree of hypoactivation in vmPFC is associated with a higher likelihood for relapse to alcohol use, which the authors and others described as a “neural marker” or “biomarker” (Blaine et al. 2015; Seo et al. 2013; Volkow and Baler 2013). Although it is true that vmPFC activity *could be a potential* biomarker—and could be used as an indicator, per the BEST workgroup (FDA-NIH Biomarker Working Group 2016; ACRI Network et al., 2024)—we believe that this finding has far to go before it is established as a biomarker, for several reasons. First, researchers must conduct rigorous tests of sensitivity and specificity, and quantify both positive and negative predictive values when applying the measure to new individual cases. We expect that vmPFC would fail in this regard, as hypoactivation of the vmPFC is not a *specific* marker of relapse risk in alcohol use disorder (e.g., hypoactivation in vmPFC is common in PTSD; Etkin and Wager 2007)—and single regions are likely too noisy to serve as neuromarkers (Chang et al. 2015). Second, the findings linking vmPFC and relapse to alcohol use represent a within-sample statistical association, which does not equate to out of sample prediction (Poldrack et al. 2020). Rather than relying on ordinary least squares regression models which can overfit the data, researchers should compare performance against null or non-informative models (e.g., predicting the mean, predicting most likely class). Third, the brain measure has not yet been used to predict outcome in new individuals, making the generalizability of the marker

in new samples unclear. Thus, there is a fundamental discrepancy in the requirements set forth by the BEST tool (FDA-NIH Biomarker Working Group 2016) and the methods typically used to identify “potential biomarkers” in the human neuroimaging literature.

A useful strategy for establishing specificity is “cross-testing” during neuromarker development. Two strategies for cross-testing neuromarkers are evaluating a putative neuromarker on data intended to elicit phenomena similar to (convergent validity) or unrelated to (divergent validity) the phenomenon of interest. For example, researchers could apply signatures/patterns developed for a particular construct (e.g., craving) to data intended to elicit similar appetitive responses (e.g., wanting/liking). Alternatively, a neuromarker intended to index craving should accurately predict self-reported craving but not predict other emotional experiences (e.g., negative affect; Koban et al. 2023b), and vice versa. To test this, researchers may apply a neuromarker to data intended to elicit experiences other than the construct of interest. Relatedly, when using visual stimuli, researchers can test whether intrinsic qualities of the stimuli map onto the neuromarker’s construct. For example, predictions from EmoNet (Kragel et al. 2019)—a deep learning neural network predictive of 20 different affective states from stimulus features—could help assess the degree to which the neuromarker’s predictions correlate with or is driven by craving-related visual features (Koban et al. 2023b).

Cross-validated prediction

Additionally, potential neuromarkers must, in fact, be predictive. Nearly half of neuroimaging studies reporting “predictive” models from 2017 to 2018 reported a statistical association as evidence for prediction (Poldrack et al. 2020). The notion that a statistical association between the brain and behavior is sufficient for prediction broadly—and to define a neuromarker specifically—is misguided for several reasons. First, the association may not be specific (e.g., hypoactivation in vmPFC; Blaine et al. 2015; Etkin and Wager 2007). Second, effect sizes for positive findings selected from a larger set of tests (e.g., voxels in a brain map) are necessarily inflated due to selection bias (Reddan et al. 2017). Conducting a set of tests and selectively presenting findings from a subset (i.e., effect size in significant voxels) is a form of overfitting (Vul et al. 2009). Third, standard regression models suffer from overfitting that results in poorer generalization of the model to new data (Copas 1983). This is because the same observations are used to estimate the model parameters (i.e., train the model) and evaluate error (i.e., test the model). Thus, random errors on observations influence both parameter estimates and predictive accuracy, leading to misestimation of both error and effect size. As a result, the training and testing errors are dependent, and

the estimated effect size from standard models is upwardly biased relative to the effect size of out-of-sample predictions.

To overcome this issue, out-of-sample cross-validation can be used to obtain unbiased estimates of effects by iteratively partitioning observations into training and testing sets, eliminating the error dependence. It is worth noting that we refer here to held-out subject data, rather than held-out data within-subjects, as this subject-level cross-validation enhances the generalizability of the resulting model to the broader population—a requirement for establishing neuromarkers. The current gold standard cross-validation schemes entail partitioning the data into several subsets (i.e., k subsets/ k folds), whereby a model is trained on a nearly full dataset, with the exclusion of a fold (subset); the fit of the model is then tested on the held-out fold. For example, tenfold cross-validation iteratively trains on 90% of the participants and tests on the held out 10%. This increases the bias towards chance performance, as a smaller subset of observations are used in training, and protects against the risk of model overfitting. At present, 5- or tenfold cross-validation are often preferred over older leave-one-out cross-validation schemes (Poldrack et al. 2020). Fortunately, there are some common rules for appropriately developing and cross-validating predictive models, reviewed in detail elsewhere (Poldrack et al. 2020; Scheinost et al. 2019; Woo et al. 2017).

Interpretability

Because machine learning algorithms make use of high dimensional datasets, interpretability can be problematic. Indeed, interpretability remains a major pitfall of the complexity of machine learning models, in which there may be thousands of features contributing to prediction. Fortunately, to improve model interpretability, there are numerous strategies available. One is data reduction methods—such as least absolute shrinkage and selection operator and ridge-regularization methods (Grosenick et al. 2013; Koban et al. 2023b) or selecting only the most relevant (i.e., correlated with behavior at some threshold) features across numerous folds of cross-validation (Shen et al. 2017)—that constrain the input of models to fewer features. However, this may result in less stable or irreproducible sets of regions, if the activation in the selected features is highly correlated. Alternatively, bootstrapping can identify peak regions with significant weights, permitting interpretation of the relationship between those voxels and the construct of interest (Koban et al. 2023a, b). Researchers may also consider selecting only features that discriminate a target state from multiple control conditions (van 't Hof et al. 2022)—or examine which voxels are associated with simple, univariate encoding and multivariate encoding, as well as the overlap between the two (Haufe et al. 2014; Wager et al. 2011). Likewise, it can

be useful to estimate structure coefficients (model encoding maps) to further enhance interpretability; whereas high model weights in a multivariate pattern/signature identify voxels important for prediction (but which are also dependent upon the covariance structure and noise in the data—i.e., brain covariates in the model), structure coefficients identify voxels individually associated with the predicted outcome (independent of brain covariates; Haufe et al. 2014). Thus, it is then possible to identify model features (here, voxels) with both high model weights and high structure coefficient values—which jointly indicate the voxels most associated with the outcome (Čeko et al. 2022). To interpret the role for specific regions or systems, researchers can use permutation testing to scramble information embedded within regions or systems (Nielsen et al. 2023) or exclude information from the region or system in analysis to assess the impact on prediction (Petre et al. 2022). Last, another strategy is to compare a neuromarker in development to resting-state networks and meta-analytic topic maps (e.g., Neurosynth; Kohoutova et al. 2020), and perhaps even use these tools to identify *a priori* regions of interest for neuromarker development (Goldstein-Piekarski et al. 2018; Wager et al. 2013).

Nuisance variables and sample size

Of course, as with any neuroimaging study, researchers must also be cognizant of the influence of nuisance variables and sample size on findings. Certainly, human neuroimaging is susceptible to influence of nuisance variables, like head motion or eye blinks. Nuisance variables systematically impact the features used for predictive modeling (e.g., activation estimates), and some of the earliest “predictive models” largely capitalized on noise (e.g., motion; Eloyan et al. 2012). Another consideration worth mentioning is sample size, as larger samples produce more reproducible brain-behavior estimates (Marek et al. 2022). That said, more modest samples remain useful for developing predictive models when using appropriate multivariate methods and cross-validation (Spisak et al. 2023). Thus, informative and interpretable neuromarkers will need robust checks against influence from noise and potential confounds, and careful attention to potential biases in results stemming from the typically small samples of many neuroimaging studies.

Validation and generalization

After developing a putative neuromarker and establishing sufficient cross-validated prediction accuracy comes yet another challenge: validation and generalization. Although cross-validation tests a model on held-out subjects, prospectively testing a model on independent and novel data is a more robust test of generalizability, and is necessary for proper validation. Certainly, neuromarkers must generalize

to independent observations, including across unique participants/individuals, multiple scanners/sites, and various research teams, tasks, and stimuli. Neuromarkers that fail to perform adequately in novel contexts will require revision and careful attention to potential biases and non-independence in development (e.g., assessing multiple model algorithms and selectively reporting the best performing model; Poldrack et al. 2020; Varoquaux and Cheplygina 2022; Woo et al. 2017). Neuromarkers which do generalize to novel contexts, though, hold promise for validation in pre-registered studies and trials, which will facilitate translation from computation to the clinic (Yip et al. 2022).

Types of neuromarkers

Susceptibility neuromarkers

With these prerequisites for neuromarker development in mind, we now consider different types of neuromarkers, each with unique associated use cases. Prior to disease onset, *susceptibility neuromarkers* identify individuals at risk for the transition from health to pathology. In SUDs, a susceptibility neuromarker would index the potential for a currently healthy individual—i.e., without an SUD—to develop a future SUD. Identifying individuals most at-risk for problematic use patterns will aid in targeting early prevention strategies and resources. Although susceptibility neuromarkers provide opportunity to intervene at the earliest stage—i.e., prior to disorder onset—they may prove most challenging to develop and validate. Indeed, we are not aware of any validated neuromarkers or biomarkers of future SUD risk in individuals without SUD. Many neurobiological changes that follow prolonged exposure to substances are a consequence of use (Berridge and Robinson 2016; Di Chiara and Imperato 1988; Koob and Volkow 2016), but susceptibility neuromarkers can only rely on pre-existing differences that predispose individuals to developing SUDs, and thus may prove more difficult to develop. Notably, one candidate biomarker includes genes that regulate aldehyde dehydrogenase, which are associated with likelihood of developing an alcohol use disorder (AUD; Tawa et al. 2016). Similarly, the additive effects across many common single nucleotide polymorphisms help to explain variance in dependence liability across a range of substances (Palmer et al. 2015), suggesting that genetic markers may be useful. However, the relationships are weak and likely complex, and may not be strong enough to be of practical significance in preventing SUDs. Still, as with other biomarkers, establishing genetic markers of SUD will require quantification of the marker’s sensitivity, specificity, positive and negative predictive values, as well as systematic tests of forward predictive utility in new individuals, prior to the onset of problematic substance use.

Imaging consortia that leverage longitudinal designs (e.g., repeatedly measuring brain responses during transitional periods from health to pathology) and aggregate data across many research sites will be crucial for developing and validating brain-based susceptibility neuromarkers. For example, the Adolescent Brain and Cognitive Development (ABCD) study, which tracks neurobiological and behavioral development from ages 9–10 into adolescence and young adulthood (Casey et al. 2018), includes the highest risk period for first alcohol and nicotine use (i.e., ages 10–16; Wittchen et al. 2008) and may yield data that could be used for the development of biomarkers and neuromarkers of substance use. In a recent effort to leverage the first-wave data from ABCD, researchers have clustered participants into distinct neurodevelopmental profiles associated with relatively low or high reward sensitivity, impulsivity, and emotion regulation prior to adolescence (Lichenstein et al. 2022)—factors that have been previously linked to SUD development (Castellanos-Ryan et al. 2013; Kober 2014). That said, these profiles were not associated with future substance use in this sample. Indeed, the approach does not yet meet any of the prerequisites outlined above for putative susceptibility neuromarkers, and instead identifies statistical associations among classes of participants which are clustered by their pattern of local brain activity across several tasks. Still, future data from ABCD or other consortia may prove useful, and further efforts to compile neuroimaging datasets from multiple sites and research teams will provide valuable opportunities for validating putative susceptibility (and other) neuromarkers across a variety of methodological parameters (Ekhtiari et al. 2022).

Diagnostic neuromarkers

Other neuromarkers inform the natural progression of a pathology. For instance, a *diagnostic neuromarker* accurately classifies individuals currently experiencing a disease or pathology, such as individuals with versus without an SUD diagnosis. Critics argue that such models may provide limited insight by simply recapitulating the extant diagnostic system (American Psychiatric Association 2013), which fails to incorporate a neurobiological basis of pathology (Hyman 2007; Kotov et al. 2017). Still, identifying neurobiological differences between individuals with versus without a diagnosis is an important benchmark of establishing meaningful neurobiological models of pathology (Woo et al. 2017). By doing so, diagnostic biomarkers provide a tool for assessing convergent validity with the clinical presentation of a disorder (e.g., self-reported symptoms), help to explain the biological contributors to disease processes (and thus potentially validate existing diagnostic categories), and provide targets for intervention.

Some researchers have built complex models sufficient for distinguishing between adolescents that do or do not misuse alcohol with reasonable accuracy (approximately 75%), although the model included a burdensome range of measures, spanning personal history (e.g., relationships, smoking status), personality (e.g., conscientiousness), genetic, brain (e.g., gray matter volume, functional activation), and performance on in-laboratory tasks (Whelan et al. 2014). Despite the inclusion of brain-based and other biological predictors in the model, and the appropriate use of cross-validation, it is, by definition, not a *biomarker*. Further, the model's predicted outcome centered on risky use rather than a diagnosis of SUD. Additionally, the complexity of the model underscores a potential challenge for biomarkers: feasibility in the clinic. That is, biomarkers comprising composites of many tasks and variables—even if all measures are biological in nature—will likely be less feasible to implement in clinical practice, to assess repeatedly as pathology progresses, or meet the stringent criteria required for biomarker validation.

More recently, we developed the neurobiological craving signature (NCS), a multivariate fMRI-based *diagnostic neuromarker* that reliably differentiated individuals with SUDs from those without SUD, based on brain activity (NCS pattern expression) during exposure to drug-related cues (Koban et al. 2023b). In one analysis using the difference between neural activity to drug cues minus neural activity to food cues, the NCS classified the two groups with 82% accuracy (sensitivity = 97%, specificity = 60%). Although classification performance remains imperfect—as expected (FDA-NIH Biomarker Working Group 2016)—performance rivals other biomarkers routinely used in clinical practice (e.g., COVID-19 antigen test sensitivity is only 65%, and even lower among asymptomatic cases; diagnostic mammograms show ~86% sensitivity and ~91% specificity; Jegerlehner et al. 2021; Lehman et al. 2015).

Prognostic/monitoring neuromarkers

Neuromarkers can also inform the progression of pathology after onset of SUD or another disorder, such as *prognostication of future SUD severity* or *monitoring of SUD severity over time*. For example, a prognostic neuromarker might identify individuals at risk for relapse to substance use following a period of abstinence (e.g., in-patient treatment). One promising finding is that drug cue-related activity in the nucleus accumbens/ventral striatum, for instance, is associated with future relapse likelihood in a sample of individuals with stimulant use disorder (MacNiven et al. 2018) and more broadly (Vafaie and Kober 2022). Still, this and other similar findings (e.g., Blaine et al. 2015; Seo et al. 2013) fail to meet the burden-of-proof required of neuromarkers (e.g., validation in out-of-sample cohorts, across substances, tasks, and research sites). In a similar manner, monitoring

neuromarkers that track SUD severity could help identify individuals at greatest need of intervention or most at risk for adverse outcomes. Perhaps unsurprisingly, greater activation along the mesolimbic dopamine pathway in response to drug cues has been cross-sectionally associated with more severe alcohol use, at least in one sample (Sjoerds et al. 2014). As such, it is possible that cue reactivity along this pathway holds promise for identifying individuals with greater sensitivity to environmental drug cues. That said, there remain no validated predictive models of addiction or drug use severity to our knowledge. By developing specific and sensitive prognostic and monitoring [severity] neuromarkers, the field will move closer to online/active assessment of disease progression both in the absence and presence of intervention.

Response neuromarkers

In the clinic, neuromarkers could inform the efficacy of treatments by examining the degree to which a particular intervention modulates the neuromarker response. In other words, assessing a neuromarker before and after an intervention could objectively index treatment response. Such measures would represent a *response neuromarker*. A subclass is a pharmacodynamic biomarker, a biomarker for drug effects on a biological process that shows evidence for penetrance into tissue and/or engagement of an intended mechanism of action. Objective measures of engagement of biological (i.e., brain) targets and treatment response in SUD are sorely needed (Volkow et al. 2015). As noted above, individuals with SUDs often suffer from impaired insight into the severity of the disorder and its symptoms (e.g., fewer than 20% seek treatment; Goldstein et al. 2009); self-report may be too inaccurate, and long-term behavior change too slow and variable to provide for the efficient evaluation of a treatment. Response neuromarkers provide an objective benchmark by which to evaluate treatment target engagement and treatment effectiveness, and can be used in conjunction with self-reported changes in symptoms (e.g., cravings, use patterns). Human neuroimaging holds promise for future development of response neuromarkers. For example, the opioid antagonist naltrexone produces concomitant decreases in fMRI measures of cue reactivity, including in the nucleus accumbens/ventral striatum among other regions, as well as self-reported desire for alcohol in response to olfactory and visual cues in patients with alcohol use disorder (Bach et al. 2020; Lukas et al. 2013; Schacht et al. 2017).

In fact, the NCS (Koban et al. 2023b)—which is freely available and ready for research use—could be evaluated as a response neuromarker in the context of pharmacological treatments known to mitigate cue reactivity in mesolimbic regions as well as self-reported cravings (Bach et al. 2020; Lukas et al. 2013). Such efforts could uncover meaningful changes in neuromarker expression that could further

understanding of treatment mechanisms, and also provide evidence of treatment effects that remain beneath conscious awareness (e.g., due to slow/incremental changes in neurobiological response or impaired insight into functional status). During its development, the NCS showed sensitivity to a cognitive regulation strategy known to reduce the intensity of self-reported cue-induced cravings (i.e., thinking about adverse consequences of smoking; Kober 2014; Kober et al. 2010a, b; Naqvi et al. 2015; Suzuki et al. 2020), that was drawn from cognitive-behavioral treatments for SUDs (Carroll 1998). Specifically, expression of the NCS pattern was lower when regulating cravings than simply looking at the images (Koban et al. 2023b). Thus, it is likely that other behavioral strategies for regulating craving and negative affect, like mindfulness-based acceptance, may similarly reduce NCS response (Kober et al. 2019; Roos et al. 2023; Westbrook et al. 2013). Future work should test this, and also consider how longer-term behavioral interventions and skills training, pharmacological, and transcranial magnetic stimulation interventions could further modulate the NCS response.

Predictive neuromarkers

Clearly, neuromarkers hold great promise for improving clinical outcomes in SUDs and beyond. One final and particularly exciting promise of neuromarker development is movement toward personalized medicine by stratifying subgroups (i.e., biotypes) of patients through *predictive neuromarkers*. It would be useful to predict which individuals respond favorably to a given treatment for several reasons. First, because the suite of treatments available to address SUDs is diverse (Carroll 2021)—spanning cognitive-behavioral (Kober et al. 2010a, b; Magill et al. 2019) and mindfulness-based interventions (Brewer et al. 2011; Li et al. 2017; Roos et al. 2023) to contingency management (Prendergast et al. 2006) and pharmacological interventions (e.g., opioid antagonists [naltrexone]; Bach et al. 2020; Lukas et al. 2013). As such, neuromarker development should include efforts to predict treatment response in a treatment-specific way. For instance, individuals with more severe SUDs tend to respond better to Mindfulness-Based Relapse Prevention than do individuals with less severe SUDs (Roos et al. 2017). Though speculative, this could be because mindfulness-based strategies, like acceptance of cravings or negative affect, do not recruit lateral prefrontal regions implicated in cognitive control (Kober et al. 2019; Westbrook et al. 2013) that are often compromised in individuals with SUDs (Goldstein et al. 2009). Notably, the NCS includes negative weights in the lateral prefrontal cortex (i.e., greater activity is associated with less craving). Thus, the NCS or other neuromarkers could, in theory, identify individuals pre-treatment that could potentially benefit more

from mindfulness-based than other cognitive-behavioral interventions that rely on effortful control and lateral PFC (Buhle et al. 2014; Kober and Mell 2015; Kober et al. 2010a, b). A similar approach could be taken for pharmacological treatment-matching as well.

Second, it would be useful to predict which individuals respond to a given psychiatric/psychological treatment because SUDs are heterogeneous (e.g., severity, substance used, treatment responsiveness, symptoms endorsed; Carroll 2021) and co-morbid (e.g., half of individuals with an SUD will qualify for a diagnosis of a different mental health disorder; NIDA 2020). Thus, individuals differ in their response to various treatments. The purpose of predictive neuromarkers is to identify individuals that would most benefit from (or potentially be harmed by) a treatment or intervention. Efforts on this front will move the field closer to personalized medicine through treatment-matching (McMahon 2014). As recommended in the BEST resource (FDA-NIH Biomarker Working Group 2016), randomized clinical trials that include individuals with a range of biomarker values undergoing treatment could identify subgroups that will benefit most from a particular therapy or intervention. Accurately predicting response prior to treatment initiation holds great promise for alleviating substantial burden to patients, providers, and society (e.g., healthcare costs, lost productivity) stemming from SUDs by expediting effective treatment delivery.

Recent developments and future directions in neuromarkers for substance use disorders

Finally, we turn to recent developments in neuromarkers for SUDs. Earlier work identified connectivity patterns predictive of cocaine (Yip et al. 2019) and opioid relapse (Lichenstein et al. 2021), and over the past year, both task-based activation and functional connectivity approaches have yielded putative neuromarkers of craving (Garrison et al. 2023; Koban et al. 2023b), as well as a putative diagnostic neuromarker of substance use (Koban et al. 2023b) and chronic cannabis use (i.e., chronic cannabis using individuals vs. non-using individuals; Kulkarni et al. 2023). See Table 1 for a summary of neuromarkers for SUD to date. Collectively, these developments represent important progress toward neuromarkers in SUDs and addiction. That said, we consider each in turn below, with a focus on adherence to the development and assessment guidelines outlined above, as well as their potential clinical utility.

The NCS represents the first developed neuromarker of food and drug craving (Koban et al. 2023b). Importantly, in its development, we carefully followed all the guidelines presented above, including using a tenfold cross-validation procedure and bootstrapped maps to identify consistent

predictors that aid interpretability. Additionally, both training and testing data came from multiple studies—representing individuals with or without alcohol, cocaine, and tobacco use disorders—and various scanners/sites; consistently, the cross-validation procedure stratified across these study-level characteristics. The diversity of both participants and study site provides varied conditions that induce machine learning algorithms to find generalizable patterns. From this rich dataset, we identified an fMRI-based multivariate pattern, or signature, in response to drug and food cue presentation that is predictive of subsequent self-reported craving intensities (Koban et al. 2023b). Of note, the model includes weights in regions well-known for their role in craving and cue reactivity, including the striatum, insula, and amygdala (Chase et al. 2011; Di Chiara and Imperato 1988; Engelmann et al. 2012; Hill-Bowen et al. 2021; Koob and Volkow 2016; Noori et al. 2016; Schacht et al. 2013), among others. Further, the pattern successfully classified individuals with versus without SUDs (82% accuracy), as well as highest versus lowest craving within individuals (81% accuracy), and showed sensitivity to a cognitive regulation strategy (Koban et al. 2023b). Thus, even in initial development, the NCS achieves (i) sensitivity and specificity (for both person-level classification and within-person craving intensities); (ii) forward predictive utility; and (iii) generalization across scanners, sites, and substance types.

Notably, with additional validation, the NCS could ultimately fulfill numerous neuromarker purposes. For instance, its ability to accurately classify individuals with versus without SUD represents promise as a *diagnostic neuromarker*. Similarly, its sensitivity to cognitive regulation strategies suggests it may also fulfill a role as a *response neuromarker*. It is also possible that the NCS might hold promise as a *prognostic* or *predictive neuromarker*. Future research should test whether the NCS might successfully predict, for example, relapse risk or other long-term outcomes. Likewise, the NCS may well be sensitive to other strategies for regulating craving, such as the mindfulness-based strategy of acceptance (Kober et al. 2019; Roos et al. 2023; Westbrook et al. 2013), or other interventions (e.g., transcranial magnetic stimulation). These directions should be explored during future validation tests of the NCS.

The other putative neuromarkers rely on functional connectivity rather than activation. Kulkarni et al. (2023) report accuracy similar to that of the NCS when classifying chronic cannabis users versus non-users (~80%; Kulkarni et al. 2023), satisfying the need for sensitivity and specificity. That said, the authors tested numerous models with various parameters and selected the best-fitting model, introducing potential bias and non-independence in model selection (Poldrack et al. 2020; Woo et al. 2017). As a result, it may be less likely to generalize in subsequent validation testing. Likewise, some top contributors to the model predictions

Table 1 Candidate neuromarkers

Candidate neuromarker	Potential Use Case(s)	N	Tested Drug Type(s)	Model selection & cross-validation (CV)	Comments	Publication
Neurobiological Craving Signature (NCS)	Diagnostic Response - cue-induced craving prediction - sensitive to cognitive regulation	99 (59 SUD, 40 controls)	Alcohol Cocaine Nicotine	LASSO-PCR Study stratified tenfold CV	Able to predict changes in craving states (e.g., trial-by-trial prediction, sensitive to regulation)	Koban et al. (2023b)
Transdiagnostic Connectome-Based Prediction of Craving	N/A Craving Prediction	274 (66 SUD, 19 obesity, 41 prenatal cocaine exposure, 148 controls)	Alcohol Cocaine (Obesity)	CPM (Ridge Regression & PCA) Group agnostic tenfold CV (100 iterations/divisions) Leave-one-group-out CV to examine influence of group	Unable to predict changes in craving across states may limit neuromarker use case applicability	Garrison et al. (2023)
Predictive Connectivity-Based Neural Signature for Chronic Cannabis Use	Diagnostic	290 (166 SUD, 124 controls)	Cannabis	L1- and L2-regularized logistic regression, L1- and L2-regularized linear support vector machine fivefold CV	Multiple models tested and the highest performing model chosen (model overfit)	Kulkarni et al. (2023)
Connectome-Based Prediction of Cocaine Abstinence	Prognostic	53 (SUD)	Cocaine	CPM (first-degree polynomial model) Leave-one-out-CV	Leave-one-out CV less preferred to other k-fold CV approaches; Monetary-incentive delay task	Yip et al. (2019)
Connectome-Based Prediction of Opioid Abstinence	Prognostic	53 (SUD)	Opioid	CPM (first-degree polynomial model) Leave-one-out-CV	Leave-one-out CV less preferred to other k-fold CV approaches; Stroop task	Lichenstein et al. (2019)

CPM Connectome-based predictive model, CV Cross-validation, LASSO-PCR Least absolute shrinkage and selection operator-principal components regression, PCA Principal components analysis, SUD Substance use disorder

(i.e., nodes with the highest degree centrality) do not map as clearly to the regions implicated in animal research and meta-analyses (Chase et al. 2011; Di Chiara and Imperato 1988; Engelmann et al. 2012; Hill-Bowen et al. 2021; Koob and Volkow 2016; Noori et al. 2016; Schacht et al. 2013).

Garrison et al. (2023) also report developing a putative neuromarker of craving. Like the NCS, Garrison et al. (2023)'s model predicts craving in response to both food and drugs (i.e., alcohol, cocaine). The authors implemented a rigorous tenfold cross validation and a leave-one-group-out cross-validation scheme to account for influences of any specific sub-sample. However, as others have noted (Konova and Schweitzer 2023), Garrison et al. (2023)'s "craving network" seemingly captures person-specific variation related to craving; that is, it predicts average craving, but not imagery-induced change in craving (i.e., from pre- to post-imagery). This limitation is crucial in that it may undermine the possibility of the model serving either *response* or *predictive neuromarker* purposes, as it may be insensitive to the effects of treatment/intervention. Further, models trained to identify individual differences (rather than states) are more likely to suffer from influences of variables which co-occur with a SUD diagnosis (e.g., anxiety, depression, BMI, fatigue, head movement).

Indeed, a major challenge—particularly for models that rely on individual differences—will be the frequent psychiatric comorbidities among individuals with SUDs. For instance, individuals with severe psychotic disorders, relative to the general population, are more than four times as likely to engage in heavy smoking, drinking, and marijuana use (Hartz et al. 2014); attention-deficit/hyperactivity disorder and other externalizing disorders are frequently seen in adolescents who misuse substances (Kroll et al. 2020); and lifetime prevalence of mood and anxiety disorders among adults with at least one SUD are 40.9% and 29.9%, respectively (Conway et al. 2006), much higher than expected in individuals with no SUDs. Given that so many individuals with SUD will qualify for the diagnosis of an additional mental health disorder (NIDA 2020), careful attention will be required in the application and interpretation of neuromarkers which might capitalize on person-level variability. Indeed, neuromarkers which rely on within-person processes specific to drug use and SUDs (e.g., the NCS; Koban et al. 2023b) will likely outperform alternative models. Ultimately, we expect that the significant comorbidities seen in SUDs will not detract from the fact that there are psychological and neurobiological commonalities among individuals with SUDs (American Psychiatric Association 2013; Volkow et al. 2019), which can be identified and leveraged as neuromarkers.

In this context, it is clear that targeting component processes during neuromarker development is of utmost importance. Because craving is a key diagnostic criterion

of SUDs that is strongly associated with relapse and future use (for a meta-analysis, see Vafaie and Kober 2022), neuromarkers that target the component process of craving, like the NCS (Koban et al. 2023b), may hold promise for diagnostic, monitoring [severity], response, and predictive purposes. Interestingly, the cannabis diagnostic neuromarker shows an association between classification confidence and self-reported cue-induced cravings (Kulkarni et al. 2023), despite not using craving ratings in model development.

As research groups continue developing putative neuromarkers, it will be important to conduct cross-marker testing—in a manner similar to neuromarker development (see above). Some neuromarkers may be developed for specific contexts (e.g., susceptibility), whereas others could fulfill multiple purposes (e.g., NCS already shows promise for diagnostic and response purposes; Koban et al. 2023b). Cross-marker testing will inform which putative neuromarkers hold the most promise for a given aspect of clinical practice. Indeed, evidence suggests that the functional networks which disambiguate individuals with a diagnosis versus individuals without a diagnosis (e.g., SUD vs. control participants) sometimes differ from the networks which predict clinical outcomes within individuals (Yip et al. 2019). Thus, a crucial direction forward for the field is identifying the relative clinical import of models that are more sensitive to individual differences (e.g., the "craving network"; Garrison et al. 2023) versus state-specific changes (e.g., intensity in cravings, like the NCS; Koban et al. 2023b).

The advances in developing neuromarkers of core processes in SUDs, like craving, can also further our theoretical understanding of SUDs and other addictive disorders. Indeed, adding to the evidence that craving in response drug, food, or other cues rely on a shared neural substrate (Volkow et al. 2017), both the NCS and "craving network" predict cravings in response to both food and drug cues (Garrison et al. 2023; Koban et al. 2023b). Still, neither marker has been tested among individuals with eating disorders or to differentiate individuals who are obese versus not-obese. Because food cue reactivity and craving are prospectively associated with eating and weight-related outcomes (Boswell and Kober 2016), as drug cues are with use and relapse (Vafaie and Kober 2022), neuromarkers of craving could inform whether neurobiological differences among individuals with versus without obesity or eating disorders resemble differences seen among substance using and non-using individuals (in response to food cues or drug cues, respectively). Relatedly, given the re-classification of "pathological gambling" to "gambling disorder" in the "addictive disorders" category in the DSM-5 (American Psychiatric Association 2013), and evidence that individuals with problematic gambling behaviors show enhanced cue reactivity relative to control participants (Goudriaan et al. 2010; Kober

et al. 2016), neuromarkers of craving could similarly inform the neurobiological basis of gambling disorder.

Clearly, the field is making impressive strides toward neuromarkers of SUDs, despite a slow start relative to other fields (Fig. 1), but important work remains. Craving—a core feature of SUD and addiction—appears a viable target for neuromarker development (Garrison et al. 2023; Koban et al. 2023b), and even markers not explicitly intended for craving appear related to the construct (Kulkarni et al. 2023). Still, only some of these markers were developed using the “best practices” outlined above (Koban et al. 2023b), and all putative neuromarkers require additional extensive testing to determine sensitivity and specificity, predictive utility, and generalization via external validation with novel participants, tasks, and research protocols. These aims will be facilitated by large-scale data sharing efforts and imaging consortia (e.g., ABCD). During this process, cross-marker testing will identify the most effective neuromarkers for a given purpose, and potentially spur translation to clinical practice (e.g., if a candidate marker shows promise as a monitoring [severity]/response neuromarker). Further, leveraging data from randomized clinical trials will strengthen our understanding of the neurobiological basis of treatments, and perhaps inform predictions of treatment effectiveness and lead to treatment-matching and personalized medicine. Future research efforts should include individuals without SUDs that struggle with cravings, including individuals with obesity, eating disorders, and gambling disorder. In all, the field is ripe with opportunities for advancing our understanding of the neurobiological basis of SUDs as well as improving clinical practice.

Funding This research was supported by National Institute on Drug Abuse R01 DA043690 (HK), T32 DA022975 (NRH).

Data Availability Data for Figure 1 were extracted from PubMed and are available here: <https://osf.io/y8u6s/>

References

- Addiction Cue-Reactivity Initiative (ACRI) Network, Sangchooli A, Zare-Bidoky M, Jouzdani AF, Schacht J, Bjork JM, et al (2024) Parameter space and potential for biomarker development in 25 years of fMRI drug cue reactivity: A systematic review. *JAMA Psychiat* 81(4):414–425
- Ahmed SH (2010) Validation crisis in animal models of drug addiction: beyond non-disordered drug use toward drug addiction. *Neurosci Biobehav Rev* 35(2):172–184
- American Psychiatric Association (2013) Diagnostic and statistical manual of mental disorders, 5th edn. American Psychiatric Association, Washington
- Bach P, Weil G, Pompili E, Hoffmann S, Hermann D, Vollstädt-Klein S et al (2020) Incubation of neural alcohol cue reactivity after withdrawal and its blockade by naltrexone. *Addict Biol* 25:e12717
- Berridge KC, Robinson TE (2016) Liking, wanting, and the incentive-sensitization theory of addiction. *Am Psychol* 71(8):670–679
- Blaine SK, Seo D & Sinha R (2015) Peripheral and prefrontal stress system markers and risk of relapse in alcoholism. *Addict Biol* (epub ahead of print)
- Boswell RG, Kober H (2016) Food cue reactivity and craving predict eating and weight gain: a meta analytic review. *Obes Rev* 17(2):159–177
- Brewer JA, Mallik S, Babuscio TA, Nich C, Johnson HE, Deleone CM et al (2011) Mindfulness training for smoking cessation: results from a randomized controlled trial. *Drug Alcohol Depend* 119(1–2):72–80
- Buhle JT, Silvers JA, Wager TD, Lopez RB, Onyemekwu C, Kober H et al (2014) Cognitive reappraisal of emotion: a meta-analysis of human neuroimaging studies. *Cereb Cortex* 24(11):2981–2990
- Califf RM (2018) Biomarker definitions and their applications. *Exp Biol Med* 243(3):213–221
- Carroll KM (2021) The profound heterogeneity of substance use disorders: implications for treatment development. *Curr Dir Psychol Sci* 30(4):358–364
- Carroll KM (1998) Therapy manuals for drug addiction, manual 1: a cognitive-behavioral approach: treating cocaine addiction. Rockville
- Casey BJ, Cannonier T, Conley MI, Cohen AO, Barch DM, Heitzeg MM et al (2018) The Adolescent Brain Cognitive Development (ABCD) study: Imaging acquisition across 21 sites. *Dev Cogn Neurosci* 32:43–54
- Castellanos-Ryan N, O’Leary-Barrett M, Sully L, Conrod P (2013) Sensitivity and specificity of a brief personality screening instrument in predicting future substance use, emotional, and behavioral problems: 18-month predictive validity of the Substance Use Risk Profile Scale. *Alcoholism* 37(S1):E281–E290
- CDC (2019) The cost of excessive alcohol use. Retrieved from <https://www.cdc.gov/alcohol/onlineMedia/infographics/cost-excessive-alcohol-use.html>
- CDC (2022) Smoking & tobacco use: fast facts and fact sheets. Retrieved from https://www.cdc.gov/tobacco/data_statistics/fact_sheets/fast_facts/index.htm
- Čeko M, Kragel PA, Woo C-W, López-Solà M, Wager TD (2022) Common and stimulus-type-specific brain representations of negative affect. *Nat Neurosci* 25(6):760–770
- Chang LJ, Gianaros PJ, Manuck SB, Krishnan A, Wager TD (2015) A sensitive and specific neural signature for picture-induced negative affect. *PLoS Biol* 13(6):e1002180
- Chase HW, Eickhoff SB, Laird AR, Hogarth L (2011) The neural basis of drug stimulus processing and craving: an activation likelihood estimation meta-analysis. *Biol Psychiat* 70(8):785–793
- Collins AG, Shenhav A (2022) Advances in modeling learning and decision-making in neuroscience. *Neuropsychopharmacology* 47:104–118
- Contreras M, Ceric F, Torrealba F (2007) Inactivation of the interoceptive insula disrupts drug craving and malaise induced by lithium. *Science* 318(5850):655–658
- Conway KP, Compton W, Stinson FS, Grant BF (2006) Lifetime comorbidity of DSM-IV mood and anxiety disorders and specific drug use disorders: results from the National Epidemiologic Survey on Alcohol and Related Conditions. *J Clin Psychiatry* 67(2):247–257
- Copas JB (1983) Regression, prediction and shrinkage. *J R Stat Soc Ser B Stat Methodol* 45(3):311–335
- Davis KD, Aghaeepour N, Ahn AH, Angst MS, Borsook D, Brenton A et al (2020) Discovery and validation of biomarkers to aid the development of safe and effective pain therapeutics: challenges and opportunities. *Nat Rev Neurol* 16(7):381–400

- Devane WA, Dysarz FR, Johnson MR, Melvin LS, Howlett AC (1988) Determination and characterization of a cannabinoid receptor in rat brain. *Mol Pharmacol* 34(5):605–613
- Devane WA, Hanuš L, Breuer A, Pertwee RG, Stevenson LA, Griffin G et al (1992) Isolation and structure of a brain constituent that binds to the cannabinoid receptor. *Science* 258(5090):1946–1949
- Di Chiara GD, Imperato A (1988) Drugs abused by humans preferentially increase synaptic dopamine concentrations in the mesolimbic system of freely moving rats. *Proc Natl Acad Sci* 85(14):5274–5278
- Dickson-Gomez J, Weeks M, Green D, Boutouis S, Galletly C, Christenson E (2022) Insurance barriers to substance use disorder treatment after passage of mental health and addiction parity laws and the affordable care act: a qualitative analysis. *Drug Alcohol Dependence Reports* 3:100051
- Ekhtiari H, Zare-Bidokhy M, Sangchooli A, Janes AC, Kaufman MJ, Oliver JA et al (2022) A methodological checklist for fMRI drug cue reactivity studies: development and expert consensus. *Nat Protoc* 17(3):567–595
- Eloyan A, Muschelli J, Nebel MB, Liu H, Han F, Zhao T et al (2012) Automated diagnoses of attention deficit hyperactive disorder using magnetic resonance imaging. *Front Syst Neurosci* 6:61
- Engelmann JM, Versace F, Robinson JD, Minnix JA, Lam CY, Cui Y et al (2012) Neural substrates of smoking cue reactivity: a meta-analysis of fMRI studies. *Neuroimage* 60:252–262
- Etkin A, Wager TD (2007) Functional neuroimaging of anxiety: a meta-analysis of emotional processing in PTSD, social anxiety disorder, and specific phobia. *Am J Psychiatry* 164(10):1476–1488
- FDA-NIH Biomarker Working Group (2016) *BEST (Biomarkers, Endpoints, and other Tools) Resource*. Retrieved from <https://www.ncbi.nlm.nih.gov/books/NBK326791/>
- Garavan H, Pankiewicz J, Bloom A, Cho JK, Sperry L, Ross TJ et al (2000) Cue-induced cocaine craving: neuroanatomical specificity for drug users and drug stimuli. *Am J Psychiatry* 157(11):1789–1798
- Garrison KA, Sinha R, Potentza MN, Gao S, Liang Q, Lacadie C, Scheinost D (2023) Transdiagnostic connectome-based prediction of craving. *Am J Psychiatry* 180(6):445–453
- Goldstein RZ, Bechara A, Garavan H, Childress AR, Paulus MP, Volkow ND (2009) The neurocircuitry of impaired insight in drug addiction. *Trends Cogn Sci* 13(9):372–380
- Goldstein-Piekarski AN, Staveland BR, Ball TM, Yesavage J, Korgaonkar MS, Williams LM (2018) Intrinsic functional connectivity predicts remission on antidepressants: a randomized controlled trial to identify clinically applicable imaging biomarkers. *Transl Psychiatry* 8(1):57
- Goudriaan AE, De Ruiter MB, Van Den Brink W, Oosterlaan J, Veltman DJ (2010) Brain activation patterns associated with cue reactivity and craving in abstinent problem gamblers, heavy smokers and healthy controls: an fMRI study. *Addict Biol* 15(4):491–503
- Grosenick L, Klingenberg B, Katovich K, Knutson B, Taylor JE (2013) Interpretable whole-brain prediction analysis with GraphNet. *Neuroimage* 72:304–321
- Hartz SM, Pato CN, Medeiros H, Cavazos-Rehg P, Sobell JL, Knowles JA et al (2014) Comorbidity of severe psychotic disorders with measures of substance use. *JAMA Psychiat* 71(3):248–254
- Haufe S, Meinecke F, Görgen K, Dähne S, Haynes J-D, Blankertz B, Bießmann F (2014) On the interpretation of weight vectors of linear models in multivariate neuroimaging. *Neuroimage* 87:96–110
- Heilig M, MacKillop J, Martinez D, Rehm J, Leggio L, Vanderschuren LJMJ (2021) Addiction as a brain disease revised: why it still matters, and the need for consilience. *Neuropsychopharmacology*. <https://doi.org/10.1038/s41386-020-00950-y>
- Herz A (1997) Endogenous opioid systems and alcohol addiction. *Psychopharmacology* 129:99–111
- Hill-Bowen LD, Riedel MC, Poudel R, Salo T, Flannery JS, Camilleri JA et al (2021) The cue-reactivity paradigm: an ensemble of networks driving attention and cognition when viewing drug and natural reward-related stimuli. *Neurosci Biobehav Rev* 130:201–213
- Hughes J, Smith T, Kosterlitz H, Fothergill LA, Morgan B, Morris H (1975) Identification of two related pentapeptides from the brain with potent opiate agonist activity. *Nature* 258(5536):577–579
- Huys QJ, Maia TV, Frank MJ (2016) Computational psychiatry as a bridge from neuroscience to clinical applications. *Nat Neurosci* 19(3):404–413
- Hyman SE (2007) Can neuroscience be integrated into the DSM-V? *Nat Rev Neurosci* 8(9):725–732
- Jasinska AJ, Stein EA, Kaiser J, Naumer MJ, Yalachkov Y (2014) Factors modulating neural reactivity to drug cues in addiction: a survey of human neuroimaging studies. *Neurosci Biobehav Rev* 38:1–16
- Jegerlehner S, Suter-Riniker F, Jent P, Bittel P, Nagler M (2021) Diagnostic accuracy of a SARS-CoV-2 rapid antigen test in real-life clinical settings. *Int J Infect Dis* 109:118–122
- Koban L, Lee S, Schelski DS, Simon M-C, Lerman C, Weber B et al (2023a) An fMRI-based brain marker of individual differences in delay discounting. *J Neurosci* 43(9):1600–1613
- Koban L, Wager TD, Kober H (2023b) A neuromarker for drug and food craving distinguishes drug users from non-users. *Nat Neurosci* 26(2):316–325
- Kober H (2014) Emotion Regulation in Substance Use Disorders. In: Gross JJ (ed) *Handbook of emotion regulation*, 2nd edn. Guilford, New York, pp 428–446
- Kober H, Mell MM (2015) Neural mechanisms underlying craving and the regulation of craving. In: Wilson SJ (ed) *Handbook on the cognitive neuroscience of addiction*. UK Wiley-Blackwell, Oxford, pp 195–218
- Kober H, Kross EF, Mischel W, Hart CL, Ochsner KN (2010a) Regulation of craving by cognitive strategies in cigarette smokers. *Drug Alcohol Depend* 106(1):52–55
- Kober H, Mende-Siedlecki P, Kross EF, Weber J, Mischel W, Hart CL, Ochsner KN (2010b) Prefrontal-striatal pathway underlies cognitive regulation of craving. *Proc Natl Acad Sci* 107(33):14811–14816
- Kober H, Lacadie CM, Wexler BE, Malison R, Sinha R, Potenza MN (2016) Brain activity during cocaine craving and gambling urges: an fMRI study. *Neuropsychopharmacology* 41:628–637
- Kober H, Buhle J, Weber J, Ochsner KN, Wager TD (2019) Let it be: mindful acceptance down-regulates pain and negative emotion. *Soc Cogn Affect Neurosci* 14(11):1147–1158
- Kohoutova L, Heo J, Cha S, Lee S, Moon T, Wager TD, Woo CW (2020) Toward a unified framework for interpreting machine-learning models in neuroimaging. *Nat Protoc* 15(4):1399–1435
- Konova AB, Schweitzer EM (2023) Decoding craving: Insights from a brain-based connectome predictive model of subjective reports. *Am J Psychiatry* 180(6):407–409
- Koob GF, Volkow ND (2016) Neurobiology of addiction: a neurocircuitry analysis. *Lancet Psychiatry* 3(8):760–773
- Kotov R, Krueger RF, Watson D, Achenbach TM, Althoff RR, Bagby RM et al (2017) The Hierarchical Taxonomy of Psychopathology (HiTOP): A dimensional alternative to traditional nosologies. *J Abnorm Psychol* 126(4):454–477
- Kragel PA, Reddan MC, LaBar KS, Wager TD (2019) Emotion schemas are embedded in the human visual system. *Sci Adv* 5(7):eaaw4358
- Kroll DS, Feldman DE, Wang S-YA, Zhang R, Manza P, Wiers CE et al (2020) The associations of comorbid substance use disorders and psychiatric conditions with adolescent brain structure and function: a review. *J Neurol Sci* 418:117099

- Kulkarni KR, Schafer M, Berner LA, Fiore VG, Heflin M, Hutchison K et al (2023) An interpretable and predictive connectivity-based neural signature for chronic cannabis use. *Biol Psychiatry Cogn Neurosci Neuroimaging* 8(3):320–330
- Le Merrer J, Becker JA, Befort K, Kieffer BL (2009) Reward processing by the opioid system in the brain. *Physiol Rev* 89(4):1379–1412
- Lehman CD, Wellman RD, Buist DS, Kerlikowske K, Tosteson AN, Miglioretti DL, Consortium, B. C. S. (2015) Diagnostic accuracy of digital screening mammography with and without computer-aided detection. *JAMA Intern Med* 175(11):1828–1837
- Leshner AI (1997) Addiction is a brain disease, and it matters. *Science* 278(5335):45–47
- Li W, Howard MO, Garland EL, McGovern P, Lazar M (2017) Mindfulness treatment for substance misuse: a systematic review and meta-analysis. *J Subst Abuse Treat* 75:62–96
- Lichenstein SD, Scheinost D, Potenza MN, Carroll KM, Yip SW (2019). *Mol Psychiatry* 26(8):4383–4393
- Lichenstein SD, Scheinost D, Potenza MN, Carroll KM, Yip SW (2021) Dissociable neural substrates of opioid and cocaine use identified via connectome-based modelling. *Mol Psychiatry* 26(8):4383–4393
- Lichenstein SD, Roos CR, Kohler R, Kiluk B, Carroll KM, Worhunsky PD et al (2022) Identification and validation of distinct latent neurodevelopmental profiles in the Adolescent Brain and Cognitive Development study. *Biol Psychiatry* 7(4):352–361
- Lovinger DM, White G, Weight FF (1989) Ethanol inhibits NMDA-activated ion current in hippocampal neurons. *Science* 243(4899):1721–1724
- Lukas SE, Lowen SB, Lindsey KP, Conn N, Tartarini W, Rodolico J et al (2013) Extended-release naltrexone (XR-NTX) attenuates brain responses to alcohol cues in alcohol-dependent volunteers: a bold fMRI study. *Neuroimage* 78:176–185
- Luo F, Li M, Florence C (2021) State-level economic costs of opioid use disorder and fatal opioid overdose—United States, 2017. *Morb Mortal Wkly Rep* 70(15):541–546
- Lüscher C, Malenka RC (2011) Drug-evoked synaptic plasticity in addiction: from molecular changes to circuit remodeling. *Neuron* 69(4):650–663
- MacNiven KH, Jensen EL, Borg N, Padula CB, Humphreys K, Knutson B (2018) Association of neural responses to drug cues with subsequent relapse to stimulant use. *JAMA Netw Open* 1(8):e186466
- Magill M, Ray L, Kiluk B, Hoadley A, Bernstein M, Tonigan JS, Carroll KM (2019) A meta-analysis of cognitive-behavioral therapy for alcohol or other drug use disorders: treatment efficacy by contrast condition. *J Consult Clin Psychol* 87(12):1093–1105
- Mameli M, Lüscher C (2011) Synaptic plasticity and addiction: learning mechanisms gone awry. *Neuropharmacology* 61(7):1052–1059
- Marek S, Tervo-Clemmens B, Calabro FJ, Montez DF, Kay BP, Hatoum AS et al (2022) Reproducible brain-wide association studies require thousands of individuals. *Nature* 603(7902):654–660
- Massaly N, Morón JA, Al-Hasani R (2016) A trigger for opioid misuse: chronic pain and stress dysregulate the mesolimbic pathway and kappa opioid system. *Front Neurosci*. <https://doi.org/10.3389/fnins.2016.00480>
- McMahon FJ (2014) Prediction of treatment outcomes in psychiatry—where do we stand? *Dialogues Clin Neurosci* 16(4):455–464
- Mihic SJ, Ye Q, Wick MJ, Koltchine VV, Krasowski MD, Finn SE et al (1997) Sites of alcohol and volatile anaesthetic action on GABAA and glycine receptors. *Nature* 389(6649):385–389
- Möhrling L, Gläscher J (2023) Prediction errors drive dynamic changes in neural patterns that guide behavior. *Cell Rep* 42(8):112931
- Mollick JA, Kober H (2020) Computational models of drug use and addiction: a review. *J Abnorm Psychol* 129(6):544–555
- Naqvi NH, Ochsner KN, Kober H, Kuerbis A, Feng T, Wall M, Morgenstern J (2015) Cognitive regulation of craving in alcohol-dependent and social drinkers. *Alcoholism* 39(2):343–349
- Nestler EJ (2005) Is there a common molecular pathway for addiction? *Nat Neurosci* 8(11):1445–1449
- Nicosia N, Pacula RL, Kilmer B, Lundberg R & Chiesa J (2009) The economic cost of methamphetamine use in the United States, 2005
- NIDA (2020) Common comorbidities with substance use disorders research report. NIH
- Nielsen AN, Kaplan S, Meyer D, Alexopoulos D, Kenley JK, Smyser TA et al (2023) Maturation of large-scale brain systems over the first month of life. *Cereb Cortex* 33(6):2788–2803
- Noori HR, Cosa Linan A, Spanagel R (2016) Largely overlapping neuronal substrates of reactivity to drug, gambling, food and sexual cues: a comprehensive meta-analysis. *Eur Neuropsychopharmacol* 26(9):1419–1430
- Palmer RH, Brick L, Nugent NR, Bidwell LC, McGeary JE, Knopik VS, Keller MC (2015) Examining the role of common genetic variants on alcohol, tobacco, cannabis and illicit drug dependence: genetics of vulnerability to drug dependence. *Addiction* 110(3):530–537
- Peters KZ, Cheer JF, Tonini R (2021) Modulating the neuromodulators: dopamine, serotonin, and the endocannabinoid system. *Trends Neurosci* 44(6):464–477
- Petre B, Kragel P, Atlas LY, Geuter S, Jepma M, Koban L et al (2022) A multistudy analysis reveals that evoked pain intensity representation is distributed across brain systems. *PLoS Biol* 20(5):e3001620
- Poldrack RA, Huckins G, Varoquaux G (2020) Establishment of best practices for evidence for prediction: a review. *JAMA Psychiat* 77(5):534–540
- Prendergast M, Podus D, Finney J, Greenwell L, Roll J (2006) Contingency management for treatment of substance use disorders: a meta-analysis. *Addiction* 101(11):1546–1560
- Reddan MC, Lindquist MA, Wager TD (2017) Effect size estimation in neuroimaging. *JAMA Psychiat* 74(3):207–208
- Ritz MC, Lamb R, Goldberg SR, Kuhar MJ (1987) Cocaine receptors on dopamine transporters are related to self-administration of cocaine. *Science* 237(4819):1219–1223
- Roos CR, Bowen S, Witkiewitz K (2017) Baseline patterns of substance use disorder severity and depression and anxiety symptoms moderate the efficacy of mindfulness-based relapse prevention. *J Consult Clin Psychol* 85(11):1041
- Roos CR, Harp N, Vafaie N, Gueorguieva R, Frankforter T, Carroll KM, Kober H (2023) Randomized trial of mindfulness- and reappraisal-based regulation of craving training among daily cigarette smokers. *Psychol Addict Behav* 37(7):829–840
- Schacht JP, Anton RF, Myrick H (2013) Functional neuroimaging studies of alcohol cue reactivity: a quantitative meta-analysis and systematic review. *Addict Biol* 18(1):121–133
- Schacht JP, Randall PK, Latham PK, Voronin KE, Book SW, Myrick H, Anton RF (2017) Predictors of naltrexone response in a randomized trial: reward-related brain activation, OPRM1 genotype, and smoking status. *Neuropsychopharmacology* 42(13):2640–2653
- Scheinost D, Noble S, Horien C, Greene AS, Lake EM, Salehi M et al (2019) Ten simple rules for predictive modeling of individual differences in neuroimaging. *Neuroimage* 193:35–45
- Schleifer L, Eldefrawi M (1974) Identification of the nicotinic and muscarinic acetylcholine receptors in subcellular fractions of mouse brain. *Neuropharmacology* 13(1):53–63
- Seo D, Lacadie CM, Tuit K, Hong KI, Constable RT, Sinha R (2013) Disrupted ventromedial prefrontal function, alcohol craving, and subsequent relapse risk. *JAMA Psychiat* 70(7):727–739

- Shen X, Finn ES, Scheinost D, Rosenberg MD, Chun MM, Papademetris X, Constable RT (2017) Using connectome-based predictive modeling to predict individual behavior from brain connectivity. *Nat Protocols* 12(3):506–518
- Simon EJ, Hiller JM, Edelman I (1973) Stereospecific binding of the potent narcotic analgesic [3H] etorphine to rat-brain homogenate. *Proc Natl Acad Sci* 70(7):1947–1949
- Sjoerds Z, van den Brink W, Beekman AT, Penninx BW, Veltman DJ (2014) Cue reactivity is associated with duration and severity of alcohol dependence: an fMRI study. *PLoS One* 9(1):e84560
- Spisak T, Bingel U, Wager TD (2023) Multivariate BWAS can be replicable with moderate sample sizes. *Nature* 615(7951):E4–E7
- Sulzer D, Sonders MS, Poulsen NW, Galli A (2005) Mechanisms of neurotransmitter release by amphetamines: a review. *Prog Neurobiol* 75(6):406–433
- Suzuki S, Mell MM, O'Malley S, Krystal J, Anticevic A, Kober H (2020) Regulation of craving and negative emotion in alcohol use disorder. *Biol Psychiatry* 5(2):239–250
- Tawa EA, Hall SD, Lohoff FW (2016) Overview of the genetics of alcohol use disorder. *Alcohol Alcohol* 51(5):507–514
- Vafaie N, Kober H (2022) Association of drug cues and craving with drug use and relapse: a systematic review and meta-analysis. *JAMA Psychiat* 79(7):641–650
- van't Hof SR, Van Oudenhove L, Janssen E, Klein S, Reddan MC, Kragel PA et al (2022) The brain activation-based sexual image classifier (BASIC): a sensitive and specific fMRI activity pattern for sexual image processing. *Cereb Cortex* 32(14):3014–3030
- Varoquaux G, Cheplygina V (2022) Machine learning for medical imaging: methodological failures and recommendations for the future. *NPJ Digital Med* 5(1):48
- Volkow ND, Baler RD (2013) Brain imaging biomarkers to predict relapse in alcohol addiction. *JAMA Psychiat* 70(7):661–663
- Volkow ND, Boyle M (2018) Neuroscience of addiction: relevance to prevention and treatment. *Am J Psychiatry* 175(8):729–740
- Volkow ND, Koob G, Baler R (2015) Biomarkers in substance use disorders. *ACS Chem Neurosci* 6(4):522–525
- Volkow ND, Wise RA, Baler R (2017) The dopamine motive system: Implications for drug and food addiction. *Nat Rev Neurosci* 18(12):741–752
- Volkow ND, Michaelides M, Baler R (2019) The neuroscience of drug reward and addiction. *Physiol Rev* 99(4):2115–2140
- Vul E, Harris C, Winkielman P, Pashler H (2009) Puzzlingly high correlations in fMRI studies of emotion, personality, and social cognition. *Perspect Psychol Sci* 4(3):274–290
- Wager TD, Atlas LY, Leotti LA, Rilling JK (2011) Predicting individual differences in placebo analgesia: contributions of brain activity during anticipation and pain experience. *J Neurosci* 31(2):439–452
- Wager TD, Atlas LY, Lindquist MA, Roy M, Woo C-W, Kross E (2013) An fMRI-based neurologic signature of physical pain. *N Engl J Med* 368(15):1388–1397
- Wang X-J, Krystal JH (2014) Computational psychiatry. *Neuron* 84(3):638–654
- Westbrook C, Creswell JD, Tabibnia G, Julson E, Kober H, Tindle HA (2013) Mindful attention reduces neural and self-reported cue-induced craving in smokers. *Soc Cogn Affect Neurosci* 8(1):73–84
- Whelan R, Watts R, Orr CA, Althoff RR, Artiges E, Banaschewski T et al (2014) Neuropsychosocial profiles of current and future adolescent alcohol misusers. *Nature* 512(7513):185–189
- Wittchen HU, Behrendt S, Höfler M, Perkonig A, Lieb R, Bühringer GE et al (2008) What are the high risk periods for incident substance use and transitions to abuse and dependence? Implications for early intervention and prevention. *Int J Methods Psychiatr Res* 17(S1):S16–S29
- Woo C-W, Chang LJ, Lindquist MA, Wager TD (2017) Building better biomarkers: brain models in translational neuroimaging. *Nat Neurosci* 20(3):365–377
- Yip SW, Scheinost D, Potenza MN, Carroll KM (2019) Connectome-based prediction of cocaine abstinence. *Am J Psychiatry* 176(2):156–164
- Yip SW, Barch DM, Chase HW, Flagel S, Huys QJ, Konova AB et al (2022) From computation to clinic. *Biol Psychiatry Global Open Sci* 3(3):319–328

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.